

Incident Report: SSH Brute Force Attack and Privelege Escalation

1. Executive Summary

This report documents a simulated cyber attack conducted in a controlled lab environment. The attack included SSH brute force attempts, successful authentication, privilege escalation to root access, and credential access via /etc/shadow file. The activity was detected and monitored using Splunk SIEM integrated with Linux audit logs.

2. Lab Environment

Component	Details
Attacker Machine	Kali Linux
Target Machine	Ubuntu Server
SIEM	Splunk Enterprise (Windows)
Log Forwarder	Splunk Universal Forwarder
Monitored Logs	auth.log, audit.log

3. Attack Overview

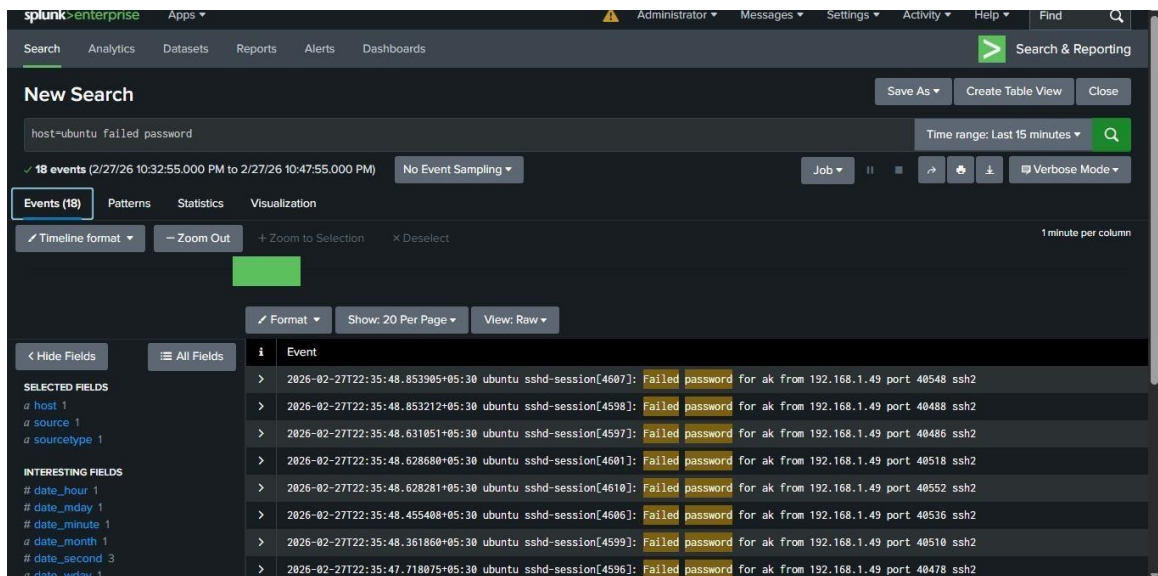
The attacker initiated a brute-force attack against the SSH service, resulting in multiple failed login attempts. After several attempts, a successful authentication was achieved using valid credentials.

Following access, the attacker escalated privileges to obtain root-level control of the system. With elevated permissions, system-level commands were executed. The sensitive file /etc/shadow was accessed, indicating potential credential compromise and full system exposure.

4. Evidence & Screenshots

4.1 Mutple Failed SSH Attempts

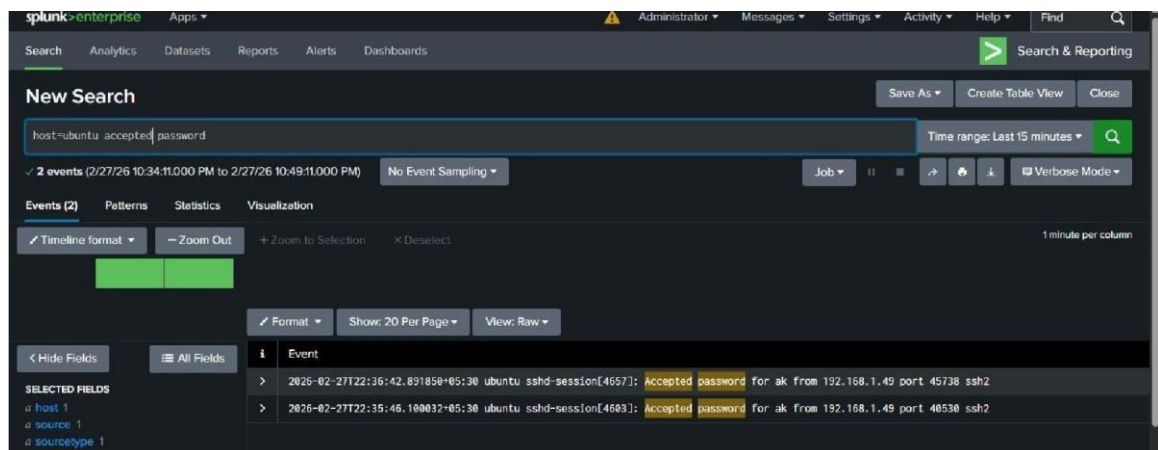
(SPL: “Failed password”)



This screenshot demonstrates repeated authentication failures from IP 192.168.1.49 indicating brute force activity.

4.2 Successful SSH Authentication

(SPL : “Accepted password”)



This event confirm the attacker successfully authenticated after multiple failed attempts.

4.3 Root-Level Command Execution

(SPL : index=* command_exec uid=0)

The screenshot shows the Splunk Enterprise interface with a search for `index=* command_exec uid=0`. The search results are displayed in a table format, showing three events. The first event is a `type=SYSCALL` message from `audit(1772212806.274:992)` with `arch=c000003e`, `syscall=59`, `success=yes`, `exit=0`, and `a0=6454120831d8`. The second event is a `type=SYSCALL` message from `audit(1772212806.269:991)` with `arch=c000003e`, `syscall=59`, `success=yes`, `exit=0`, and `a0=5d16098cf410`. The third event is a `type=SYSCALL` message from `audit(1772212806.239:990)` with `arch=c000003e`, `syscall=59`, `success=yes`, `exit=0`, and `a0=7ffc3c285680`. The events are filtered by `uid=0` and `command_exec`.

This screenshot shows execution of commands with UID=0, Confirming privilege escalation.

4.4 Access to /etc/shadow

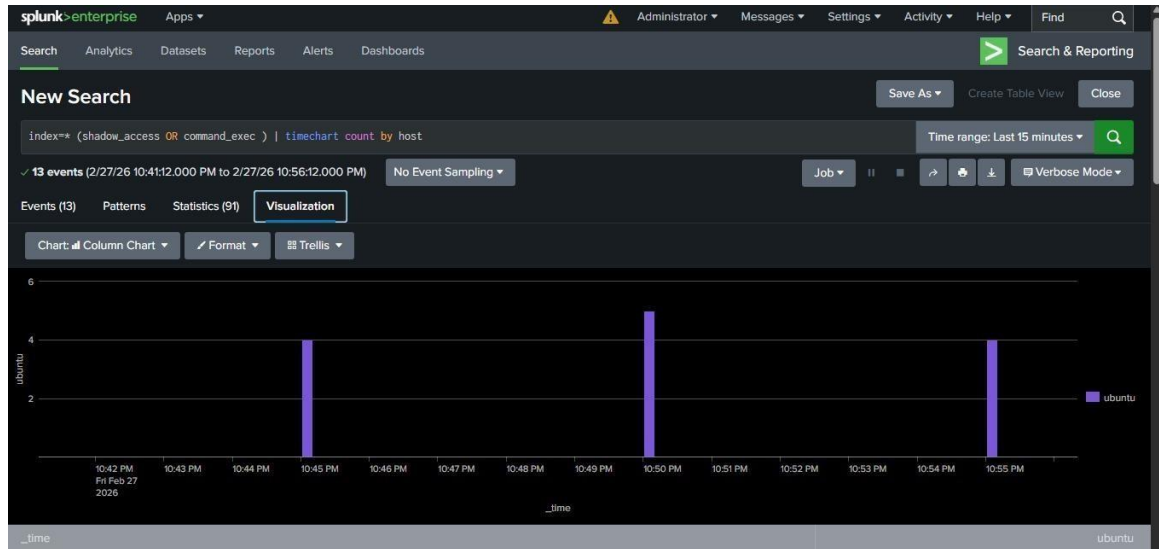
(SPL : index=* shadow_access)

The screenshot shows the Splunk Enterprise interface with a search for `index=* shadow_access`. The search results are displayed in a table format, showing one event. The event is a `type=SYSCALL` message from `audit(1772212501.689:978)` with `arch=c000003e`, `syscall=257`, `success=yes`, `exit=3`, and `a0=ffffff9c`. The event is filtered by `shadow_access`.

This event confirms access to a highly sensitive credential file.

4.5 Attack Timeline Visualization

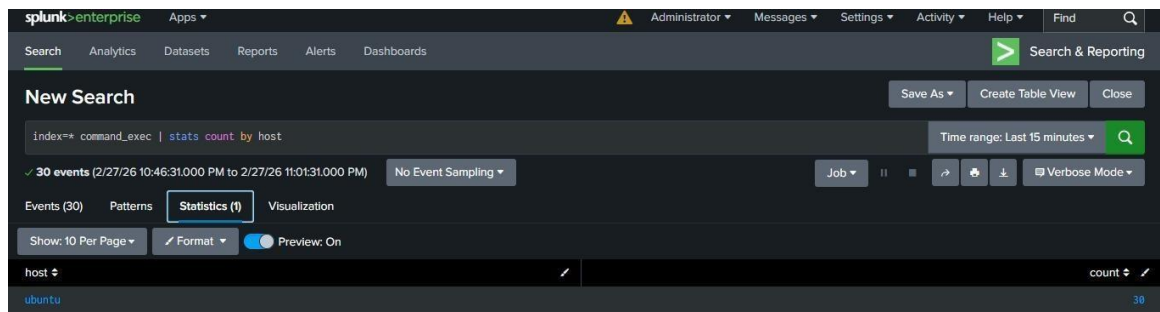
(SPL : index=* (shadow_access OR Command_exec) | timechart count by host)



The Visualization clearly shows escalation of activity over time.

4.6 Statistical Summary of Commands

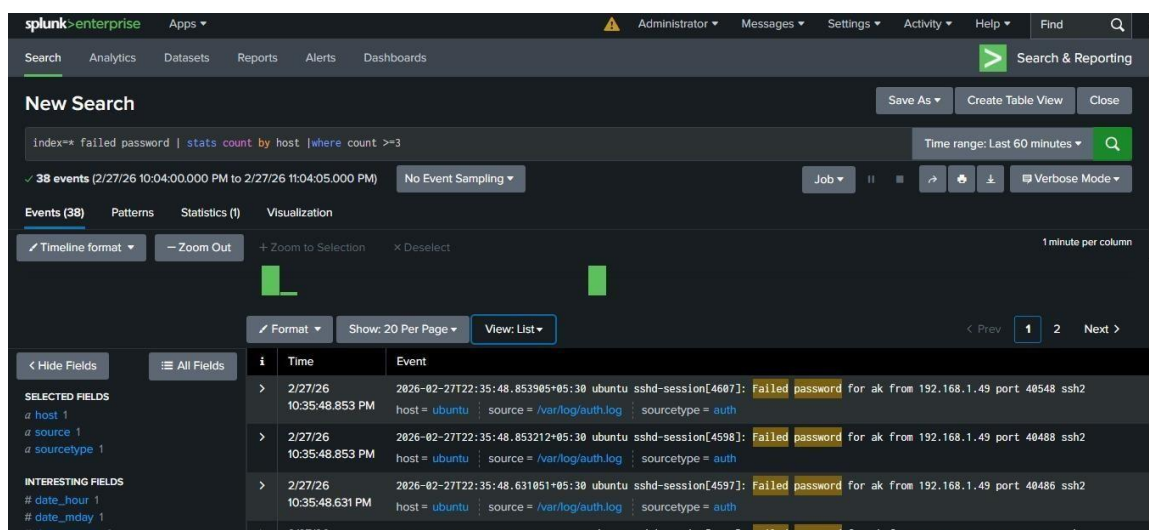
(SPL : index=* command_exec | stats count by host)



Provides a summarized view of all executed commands.

4.7 Brute Force Detection Rule

(SPL : index=* “Failed password” | stats count by host | where count >=3)



This rule triggers when 3 or more failed SSH login attempts are detected.

5.MIRTE ATT&CK Mapping

Attack Stage	Technique ID
Brute Force	T1110
Valid Accounts	T1078
Privilege Escalation	T1548
OS Credential Dumping	T1003, .008
Command Execution	T1059

6.Cyber Kill Chain Mapping

Kill Chain Phase	Description in This Incident
Reconnaissance	SSH service (Port 22) was identified as publically accessible on the target system.
Weaponization	Hydra was configured with credential wordlists to automate a brute force attack.
Delivery	Multiple login attempts were sent to SSH service to gain unauthorized access.
Exploitation	Valid credential were discovered,resulting in successful authentication.
Installation	Weak file permissions were exploited to escalate privileges within the system.
Command & Control	Root-level access was obtained,enabling full administrative control.
Actions on Objectives	The /etc/shadow file was accessed,indicating credential compromise.

7. Impact Assessment

- Root-level compromise achieved
- Credential file accessed
- High risk of lateral movement
- Potential data exfiltration risk

Severity Level : HIGH

8. Remediation Recommendations

Immediate Actions

- Change default SSH port
- Remove unnecessary sudo privileges
- Disable compromised accounts
- Disable password authentication
- Use key-based authentication
- Restrict via firewall
- Enable Fail2Ban

9. Conclusion

The simulated attack successfully demonstrated the detection of a complete attack lifecycle, beginning with SSH brute-force attempts and progressing to credential access and privilege escalation. Splunk SIEM effectively monitored, logged, and visualized malicious activities, enabling timely investigation and response.

This lab validates the effectiveness of centralized log monitoring, detection rule configuration, and auditd integration in identifying unauthorized access and privilege escalation attempts, thereby strengthening overall security posture.